



UNIVERSITÀ
DI TRENTO

Introduzione alla Programmazione per il Web

a.a. 2025-2026

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Welcome to Introduzione alla Programmazione Web!

- Who I am:
Giovanna Varni, Associate Professor @DISI
Office: 208, Povo 1
Office hours:
 - Wednesday from 1PM to 2PM (from 11/03)
 - Book your 15 minutes slot here:

[Google Calendar](#)

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Welcome to Introduzione alla Programmazione Web!

- Teaching Assistants:

Vincenzo Marco De Luca e Samuele Bortolotti

Office 173 and 196, Povo 2

Office hours: Thursday from 1PM to 2PM

Email: vincenzomarco.deluca@unitn.it

samuele.bortolotti@unitn.it

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Preliminary Schedule

FEBBRAIO 2026 FEBRUARY 2026	MARZO 2026 MARCH 2026	APRILE 2026 APRIL 2026	MAGGIO 2026 MAY 2026	GIUGNO 2026 JUNE 2026
1 D Su	1 D Su	1 M We	1 V Fr	1 L Mo
2 L Mo	2 L Mo	2 G Th	2 S Sa	2 M Tu
3 M Tu	3 M Tu	3 V Fr	3 D Sa	3 M We
4 M We	4 M We	4 S Sa	4 L Mo	4 G Th
5 G Th	5 G Th	5 D Su	5 M Tu	5 V Fr
6 V Fr	6 V Fr	6 L Mo	6 M We	6 S Sa
7 S Sa	7 S Sa	7 M Tu	7 G Th	7 D Su
8 D Su	8 D Su	8 M We	8 V Fr	8 L Mo
9 L Mo	9 L Mo	9 G Th	9 S Sa	9 M Tu
10 M Tu	10 M Tu	10 V Fr	10 D Sa	10 M We
11 M We	11 M We	11 S Sa	11 L Mo	11 G Th
12 G Th	12 G Th	12 D Su	12 M Tu	12 V Fr
13 V Fr	13 V Fr	13 L Mo	13 M We	13 S Sa
14 S Sa	14 S Sa	14 M Tu	14 G Th	14 D Su
15 D Su	15 D Su	15 M We	15 V Fr	15 L Mo
16 L Mo	16 L Mo	16 G Th	16 S Sa	16 M Tu
17 M Tu	17 M Tu	17 V Fr	17 D Sa	17 M We
18 M We	18 M We	18 S Sa	18 L Mo	18 G Th
19 G Th	19 G Th	19 D Su	19 M Tu	19 V Fr
20 V Fr	20 V Fr	20 L Mo	20 M We	20 S Sa
21 S Sa	21 S Sa	21 M Tu	21 G Th	21 D Su
22 D Su	22 D Su	22 M We	22 V Fr	22 L Mo
23 L Mo	23 L Mo	23 G Th	23 S Sa	23 M Tu
24 M Tu	24 M Tu	24 V Fr	24 D Sa	24 M We
25 M We	25 M We	25 S Sa	25 L Mo	25 G Th
26 G Th	26 G Th	26 D Su	26 M Tu	26 V Fr
27 V Fr	27 V Fr	27 L Mo	27 M We	27 S Sa
28 S Sa	28 S Sa	28 M Tu	28 G Th	28 D Su
	29 D Su	29 M We	29 V Fr	29 L Mo
	30 L Mo	30 G Th	30 S Sa	30 M Tu
	31 M Tu		31 D Su	

Course schedule:

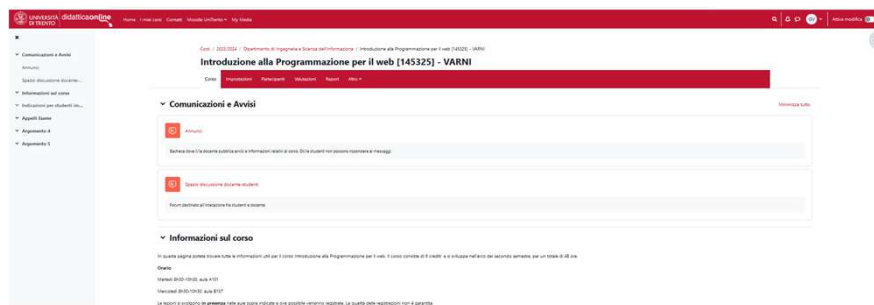
Thursday, 8:30 – 10:30

Wednesday, 8:30 – 10:30

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Moodle

- Check out the Moodle of the course for news and materials!



- If you do not have access yet, please:
 - verify with the student office
 - if the problem persists send me an email

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Pre-requirements

- Programming 1 and OOP: essential!
- Something about Networks
- Something about Databases
- No need to have previous knowledge on web

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What are we going to do?



Developing web apps!

“A **web application** (or **web app**) is application software that is accessed using a web browser. Web applications are delivered on the World Wide Web to users with an active network connection” (source: Wikipedia)

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Why is web programming complex?

- So many reasons!
- Do you know some of them?

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Methods

- In this **introductory** course we will try to address some of these major issues

HOW?

- Building a notional machine for web programming
- We will learn to use a popular Java-based framework (Spring) and other web technologies
- We will inspect and discuss together snippets of code

Let's take a look at the syllabus of the course!

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Materials, Books and References

- You can find the slides and the code showed during lectures on Moodle
- W3C schools for tutorials and gyms
- Spring tutorial (<https://spring.io/projects>)
- Some useful books:
 - Spica, L., Spring Start Here. Learn what you need and learn it well. Manning (2021)
 - Spica L., Spring Security in Action. Manning (2024)



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Exam

- 5 calls per year (June, July, September, January, February)
- You must register on ESSE3!
- Exam structure:
 - 50% Quiz and questions on course' topics: 9/16 to pass
 - 50% Project of a web app on an assigned topic: 9/16 to pass

Everything must be done on the same call

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Exam

- About project:
 - You have to submit it via Moodle **three days** before the quiz
 - Once the project is submitted:
 - You can attend the quiz for that call
 - If the project is scored less than 9 points, you can resubmit it at each call until it gets 9 or more points
 - If the project is scored more than 9 points, you can register your grade if the grade of the quiz was scored more than 9 points, or, to improve your grade, you can attend next call of the quiz, and/or resubmit once the project
 - The project must be submitted at each call you attend (also if no changes are applied)
 - The project expires with the call of February 2027

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Questions? Ready?

Go!



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Q

**What is the difference between the
Web and the Internet?**

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Let's see!

- "The Internet is a global system of interconnected computer networks that interchange data by packet switching using the standardized Internet Protocol Suite (TCP/IP)."
- Thus, the Internet is a network of networks, defined by the TPC/IP standards (such as FTP, Telnet, SMTP...).

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What is the web

The Wold Wide Web is an **information system** where documents and other web resources are identified by **Uniform Resource Locators** (URLs, such as <https://www.example.com/>), which may be interlinked by **hypertext**, and are accessible over the Internet.

The resources of the WWW are:

- described via the **Hypertext Markup Language (HTML)**
- transferred via the **Hypertext Transfer Protocol (HTTP)**
- accessed by users by a software app called a **web browser**
- published by a software app called a **web server**

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History of the web

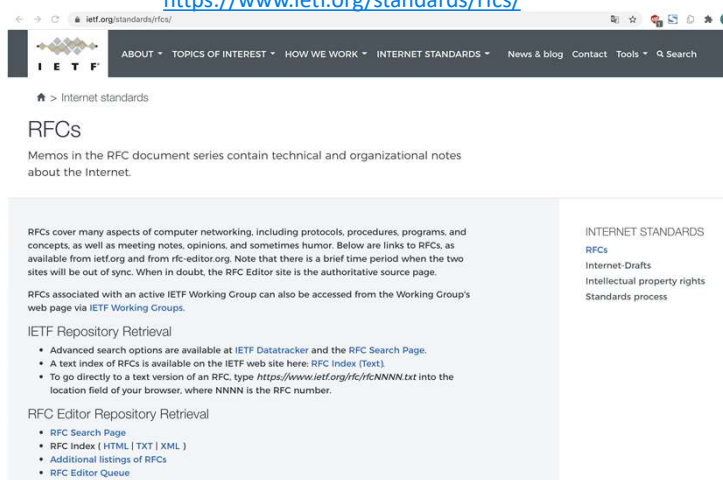
- 1989-1990 - **Tim Berners-Lee** invents World Wide Web at CERN
- 1993 - CERN puts the World Wide Web software in the public domain. Later, CERN made a release available with an open license, a more sure way to maximize its dissemination.
- 1994 - Berners-Lee moved from CERN to the Massachusetts Institute of Technology to found the **World Wide Web Consortium (W3C)**, an international community devoted to developing open web standards.

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Important references: IETF

The **Internet Engineering Task Force (IETF)** is a large open international community of network designers, operators, vendors, and researchers concerned with **the evolution of the Internet architecture and the smooth operation of the Internet.**

<https://www.ietf.org/standards/rfcs/>



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Important references: W3C Standards



Web Design and Applications

Web Design and Applications involve the standards for building and Rendering Web pages, including HTML, CSS, SVG, Ajax, and other technologies for Web Applications ("WebApps"). This section also includes information on how to make pages accessible to people with disabilities (WCAG), to internationalize them, and make them work on mobile devices.



Web of Devices

W3C is focusing on technologies to enable Web access anywhere, anytime, using any device. This includes Web access from mobile phones and other mobile devices as well as use of Web technology in consumer electronics, printers, interactive television, and even automobiles.



Web Architecture

Web Architecture focuses on the foundation technologies and principles which sustain the Web, including URIs and HTTP.



Semantic Web

In addition to the classic "Web of documents" W3C is helping to build a technology stack to support a "Web of data," the sort of data you find in databases. The ultimate goal of the Web of data is to enable computers to do more useful work and to develop systems that can support trusted interactions over the network. The term "Semantic Web" refers to W3C's vision of the Web of linked data. Semantic Web technologies enable people to create data stores on the Web, build vocabularies, and write rules for handling data. Linked data are empowered by technologies such as RDF, SPARQL, OWL, and SKOS.



XML Technology

XML Technologies including XML, XML Namespaces, XML Schema, XSLT, Efficient XML Interchange (EXI), and other related standards.



Web of Services

Web of Services refers to message-based design frequently found on the Web and in enterprise software. The Web of Services is based on technologies such as HTTP, XML, SOAP, WSDL, SPARQL, and others.



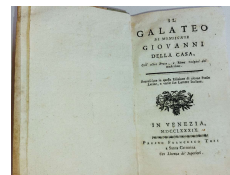
Browsers and Authoring Tools

The web's usefulness and growth depends on its universality. We should be able to publish regardless of the software we use, the computer we have, the language we speak, whether we are wired or wireless, regardless of our sensory or interaction modes. We should be able to access the web from any kind of hardware that can connect to the Internet – stationary or mobile, small or large. W3C facilitates this listening and blending via international web standards. These standards ensure that all the crazy brilliance continues to improve a web that is open to us all.

A quick overview of the basics

Protocol

- **Protocol is a synonymous of Etiquette**, a code of behavior that delineates expectations for social behavior according to contemporary conventional norms within a society, social class, or group.
- **Communications protocol**, a set of rules and regulations that determine how data is transmitted in telecommunications and computer networking



<https://www.salute.gov.it/portale/nuovocoronavirus/dettaglioNotizieNuovoCoronavirus.jsp?id=4696&lingua=italiano&menu=notizie&p=dalministero>

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Protocol

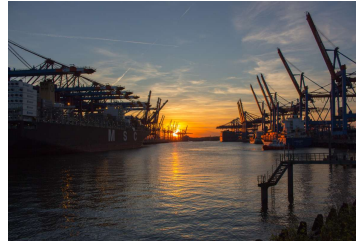
- A **protocol** defines:
 - the **format**
 - the **order** of messages exchanged between two or more communicating entities
 - the **actions** taken on the transmission and/or receipt of a message or other event
- The protocol of the Web is the HTTP protocol
HTTP: hypertext transfer protocol

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Socket and Port

A **process** associates its input or output channels with a **socket**. This procedure is known as binding.

socket: {protocol, local address, local port, remote address, remote port}



HTTP on port 80
 • HTTP with SSL (HTTPS) on port 443
 • FTP on port 21
 • SMTP on port 25
 • POP on port 110
 • SSH on port 22

PID	PORT	IP	Protocol
84	21	193.205.196.130	FTP
78	80	193.205.196.130	HTTP
321	8080	193.205.196.130	HTTP
541	25	193.205.196.130	SMTP

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URI, URL, URN

A **web resource**, or simply **resource**, is any identifiable thing, whether digital or physical

- A **Uniform Resource Identifier (URI)** is a compact sequence of characters that identifies an abstract or physical resource. (RFC 3986)
- A **Uniform Resource Locator (URL)** refers to the subset of URI that identify resources via a **representation of their primary access mechanism** (e.g., their network "location")
- **Uniform Resource Name (URN)** refers to the subset of URI that are required to **remain globally unique and persistent even when the resource ceases to exist** or becomes unavailable. It is intended to serve as persistent, location-independent, resource identifier (RFC 2141).

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URI, URL, URN

Both URL and URN are URI

- A URN identifies a resource
- A URL provides a method for finding it
- A URN may be compared to a person's tax id code
- A URL may be compared to their street address
- A URN can be associated to many URLs

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RFC

[\[RFC Home\]](#) [\[TEXT\]](#) [\[PDF\]](#) [\[HTML\]](#) [\[Tracker\]](#) [\[IPR\]](#)

PROPOSED STANDARD

Internet Engineering Task Force (IETF)

P. Saint-Andre

Request for Comments: 8141

Filament

Obsoletes: 2141, 3406

J. Klensin

Category: Standards Track

April 2017

ISSN: 2070-1721

Uniform Resource Names (URNs)

Abstract

A Uniform Resource Name (URN) is a Uniform Resource Identifier (URI) that is assigned under the "urn" URI scheme and a particular URN namespace, with the intent that the URN will be a persistent, location-independent resource identifier. With regard to URN syntax, this document defines the canonical syntax for URNs (in a way that is consistent with URI syntax), specifies methods for determining URN-equivalence, and discusses URI conformance. With regard to URN namespaces, this document specifies a method for defining a URN namespace and associating it with a namespace identifier, and it describes procedures for registering namespace identifiers with the Internet Assigned Numbers Authority (IANA). This document obsoletes both RFCs 2141 and 3406.

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URN + URL example

- The ISBN system (namespace) uniquely identifies books
- **urn:isbn:0-486-27557-4** cites unambiguously a *specific edition* of Shakespeare's play Romeo and Juliet.
- A typical URL for this book might look like the file path **file:///home/username/RomeoAndJuliet.pdf**



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URI Schemes

<scheme>: <scheme-specific-part>

http:
https:
ftp:

mailto:<address>[?<header1>=<value1>[&<header2>=<value2>]]

geo:<lat>,<lon>

fax:<phone number>

file:[//host]/path

bitcoin:<address>[?[amount=<size>]...

skype:<username|phonenumber>...

https://en.wikipedia.org/wiki/List_of_URI_schemes

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URI structure

URI = scheme:[//authority]path[?query][#fragment]

authority = [userinfo@]host[:port]

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MIME type

URIs give us info about name and location, but what about their content type?

File extensions are bad...

(see Unix magic numbers

<https://www.geeksforgeeks.org/working-with-magic-numbers-in-linux/>)

Metadata?

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MIME type

Multipurpose Internet Mail Extensions (RFC 2045, 2046)

MEDIA_TYPE/SUBTYPE

text -> text/plain, text/html, text/richtext ...

image -> image/jpeg, image/png, image/svg+xml...

audio -> audio/basic, audio/ogg, audio/x-wav...

video -> video/mp4, video/ogg...

application -> application/x-apple-diskimage... ..

multipart

See <https://www.freeformatter.com/mime-types-list.html>

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Client and Server

from “Computer Networking: A Top-Down Approach”, Kurose-Ross:

- The computers and other devices connected to the Internet are often referred to as **end systems (hosts)**.
- Hosts are sometimes further divided into two categories: **clients** and **servers**.
- Informally, clients tend to be desktop and mobile PCs, smartphones, and so on, whereas servers tend to be more powerful machines that store and distribute Web pages, stream video, relay e-mail, and so on.

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Servers ?



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Server = Waiter, Client = Customer



Server and client are SOFTWARE roles,
not HARDWARE concept!

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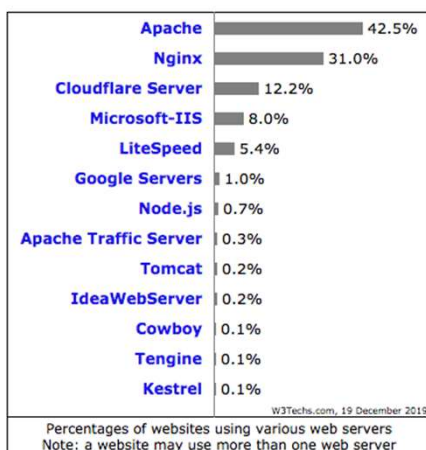
Client and Server

So, a better definition:

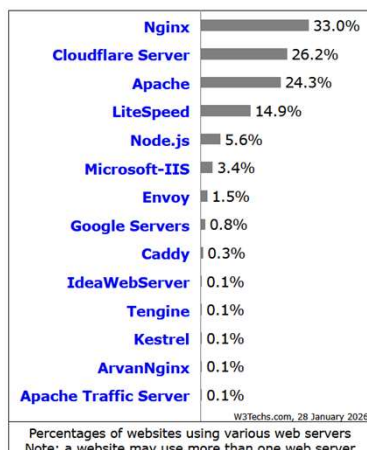
- A Server is a machine that opens a connection, and waits for incoming calls (in order to provide a service)
- A Client is a machine the starts a connection (opening a socket to the server) and requests a service.

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Servers on the web



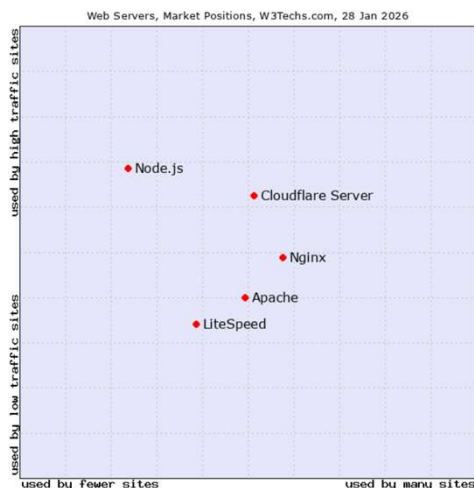
December 2019



January 2026

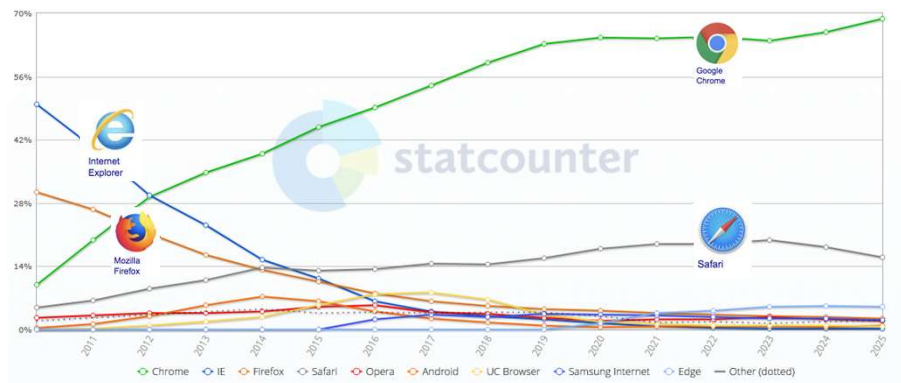
37

Servers on the web



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Clients on the Web: Browsers



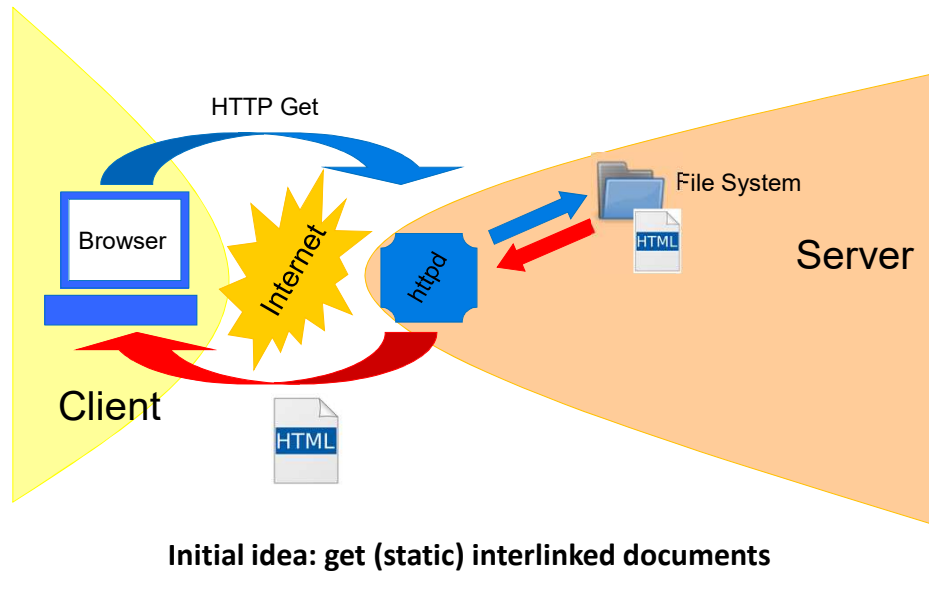
<https://gs.statcounter.com/browser-market-share#yearly-2010-2026>

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Evolution of web architecture

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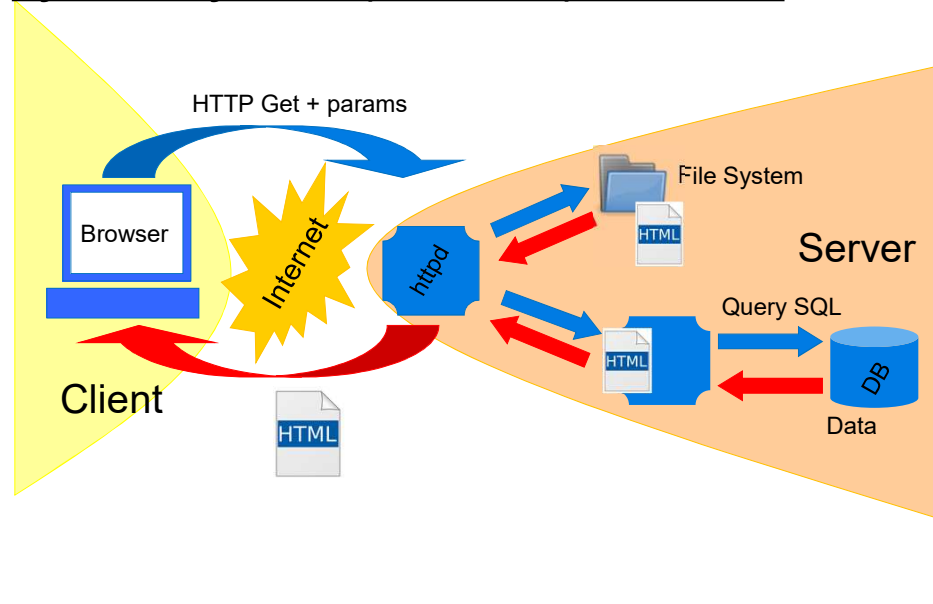
The original web architecture: static pages



The model

- Mapping 1:1 between URLs and (static) resources
- The Web server is nothing but a retriever of the content associated to an URL (typically HTML page, image...)
- The role of a web programmer:
writing collections of HTML pages

Evolution 1: Dynamically create (interlinked) documents



The model

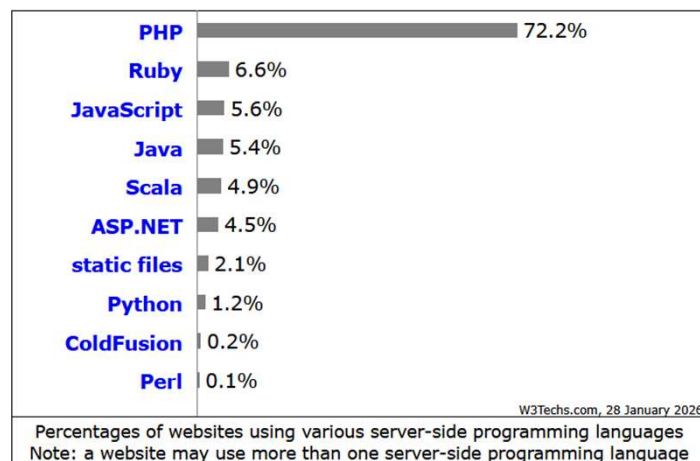
- Certain URLs are associated with **actions** and carry **parameters** for the action
- The Web server:
 - understands that certain URLs are *dynamic*
 - parses the parameters
 - starts a process (or thread) corresponding to the desired action
 - performs some computations
 - passes the results of computations to the client as html pages

The model

- The role of a web programmer:
They write collections of HTML pages and they also write programs, using the programming language of their choice

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Languages



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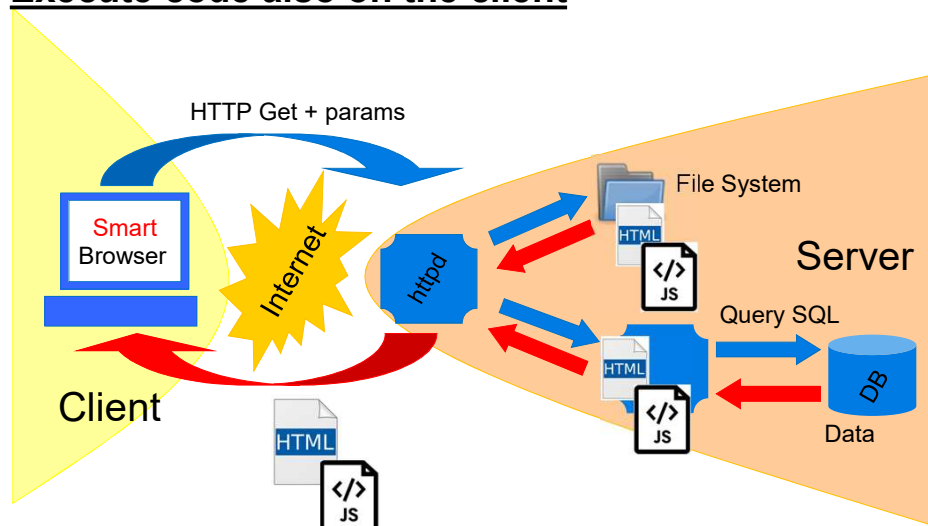
Back-end (Server-side) table in most popular websites

Websites	C#	C	C++	D	Elixir	Erlang	Go	Hack	Haskell	Java	JavaScript	Perl	PHP	Python	Ruby
Google	No	Yes	Yes	No	No	No	Yes	No	No	Yes	Yes	No	No	Yes	No
Facebook	No	No	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	No	Yes	Yes	No
YouTube	No	Yes	Yes	No	No	No	Yes	No	No	Yes	No	No	No	Yes	No
Yahoo	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
Etsy	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
Amazon	No	No	Yes	No	No	No	No	No	No	Yes	No	Yes	No	No	No
Wikipedia	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
Fandom	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
X	No	No	Yes	No	No	No	No	No	No	Yes	No	No	No	No	Yes
Bing	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No
eBay	No	No	No	No	No	No	No	No	No	Yes	Yes	No	No	No	No
MSN	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No
LinkedIn	No	No	No	No	No	No	No	No	No	Yes	Yes	No	No	No	No
Pinterest	No	No	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No
WordPress.com	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
Netflix	No	No	No	No	No	No	No	No	No	Yes	No	No	No	Yes	No

source: Wikipedia

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Evolution 2: Execute code also on the client

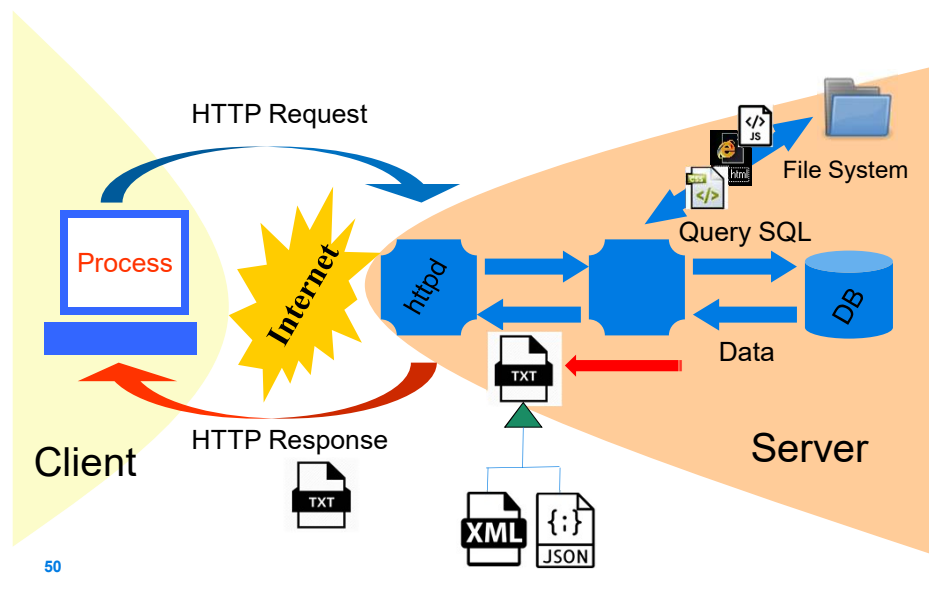


The model

- The role of a web programmer:
They also writes programs which run on the browser
- Which language?

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Evolution 3: Responsivness



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Focus on the server-side

- Two options:
 - **Single process (eventually multi-thread):**
All the back-end functionalities (e.g. database, authentication) are tightly integrated and deployed as a single unit

PROs

Simplicity
Easy Deployment
Easy Testing
Lower Overhead

CONs

Scalability
Maintenance
Flexibility
Robustness
Team Autonomy

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Focus on the server-side

- Two options:
 - **Service Oriented Architecture (SOA):**
back-end functionalities are implemented in multiple independent processes (called services) that communicate with each other over a network

PROs

Scalability
Flexibility
Maintenance
Robustness
Team Autonomy
Integration

CONs

Complexity
High Overhead
Security
Data Persistence
Testing and deployment Complexity
High Resource Requirement

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Q

Which option should I choose?

It depends on the task at hand

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